

4688_windows_security.log.

When I added the file into Splunk, it did not recognize the Windows XML format correctly. The entire log was grouped together as one event.

view Event Summary

	Time	Event
1	6/23/26 10:12:15.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-A5BA-3E3B0328C30D}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2022-08-31T17:47:04.788786400Z' /><EventRecordID>31213331</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='816' /><Channel>Security</Channel><Computer>win-dc-mhaag-attack-range-602.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0xd08f94</Data><Data Name='NewProcessId'>0x18bc</Data><Data Name='NewProcessName'>C:\Windows\System32\ntdsutil.exe</Data><Data Name='TokenElevationType'>%%1936</Data><Data Name='ProcessId'>0x90c</Data><Data Name='CommandLine'>"C:\Windows\system32\ntdsutil.exe" "ac i ntds" ifm "create full C:\Windows\Temp\ntds_T1003" q q</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserName'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event> <Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-A5BA-3E3B0328C30D}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2022-08-31T17:46:58.657885900Z' /><EventRecordID>31213313</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='816' /><Channel>Security</Channel><Computer>win-dc-mhaag-attack-range-602.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0xd08f94</Data><Data Name='NewProcessId'>0x1a84</Data><Data Name='NewProcessName'>C:\Windows\System32\ntdsutil.exe</Data><Data Name='TokenElevationType'>%%1936</Data><Data Name='ProcessId'>0xec0</Data><Data Name='CommandLine'>ntdsutil "ac i ntds" "ifm" "create full C:\Windows\Temp\ntds_T1003" q q</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserName'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event>

To parse the logs correctly, add a custom source type and in the Advanced section add the following rules:

Name	Value	
LINE_BREAKER	([\r\n]+)(?=<Event)	×
SHOULD_LINEMERGE	false	×
category	Custom	×
pulldown_type	true	×
TRUNCATE	0	×
KV_MODE	XML	×

New setting

Copy to clipboard

Apply settings

SHOULD_LINEMERGE = false

By running this query, we confirmed that all 8 events were EventCode 4688.

Events	Patterns	Statistics (1)	Visualization
Show: 20 Per Page ▾	Format ▾	Preview: On	
EventCode ↕			count ↕
4688			8

We know that to find evidence of T1003.003, we should look for the creation of processes such as:

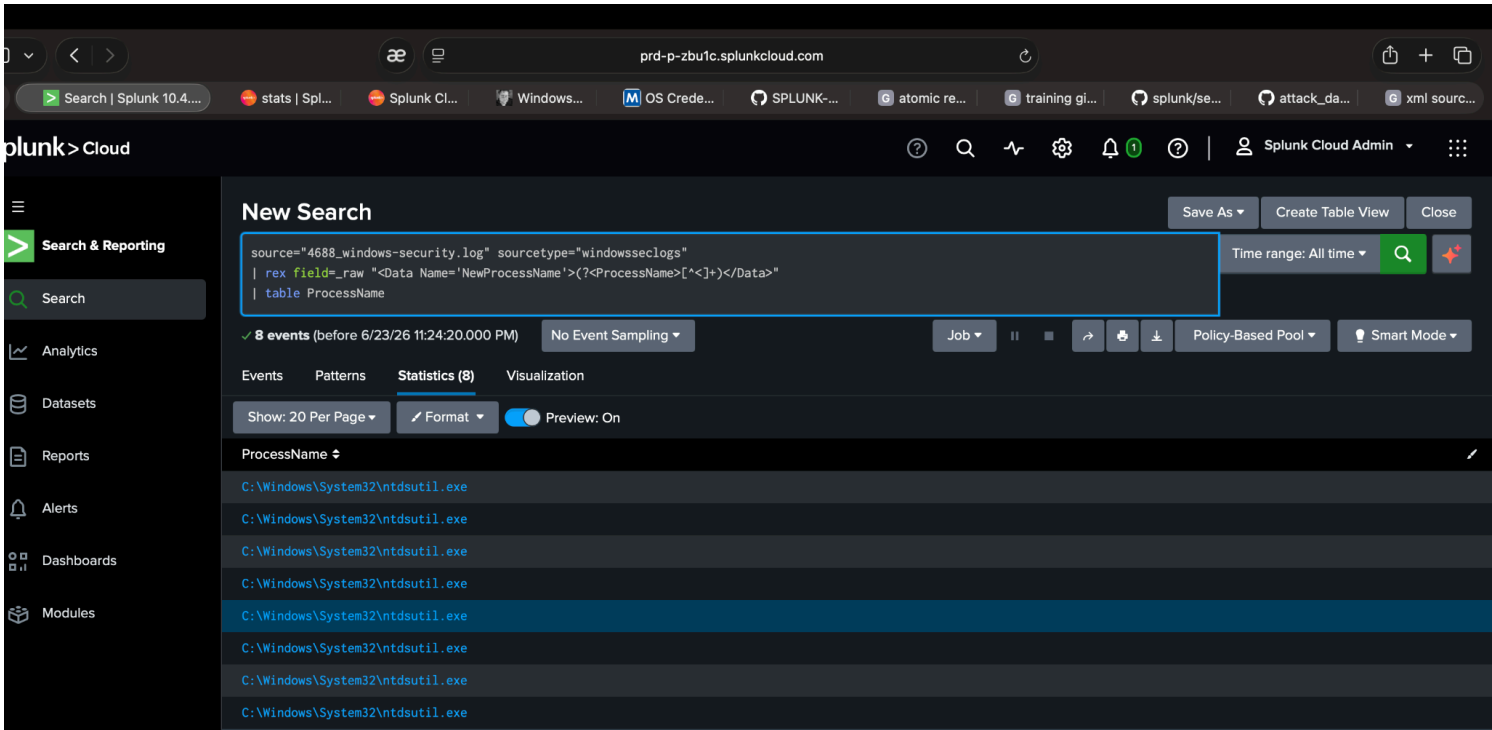
Event ID	Log Source	What to Look For	Suspicious Process
4688	Windows Security	New process created	ntdsutil.exe
4688	Windows Security	Shadow copy creation commands	vssadmin.exe
4688	Windows Security	Shadow copy commands	wmic.exe
4688	Windows Security	Disk shadow operations	diskshadow.exe
4688	Windows Security	Database copy activity	esentutl.exe
1	Sysmon Process Create	Same commands with full command line	Any of the above
11	Sysmon File Create	Creation or copy of .dit files	Any process
4656	Security Object Access	Read access to NTDS.dit	Any process
4663	Security Object Access	File access to NTDS.dit	Any process
8222	VSS Service	Volume Shadow Copy created	Usually followed by copy commands

ntdsutil.exe is a normal Windows administration tool used on Domain Controllers to manage Active Directory. Attackers abuse it in T1003.003 to copy the Active Directory database and steal password hashes.

We could search directly for ntdsutil.exe:

```
source="4688_windows-security.log" sourcetype="windowsseclogs"
| rex field=_raw "(?[^<]+)"
| search ProcessName="ntdsutil"
```

Evidence 1



All 8 events had EventCode 4688 and showed the creation of:

C:\Windows\System32\ntdsutil.exe

The parent process responsible for the process creation was cmd.exe in 7 events and powershell.exe in 1 event.

The executing user was Administrator.

Evidence 2

Observation	Details
Time Range	All 8 executions occurred between 17:34 and 17:47 (13-minute window)
Frequency	Multiple executions occurred within seconds of each other
Pattern	Example: Events at 17:38:40 and 17:38:52 occurred only 12 seconds apart

New Search

Save As ▾

Create Table ▾

```
source="4688_windows-security.log" sourcetype="windowsseclogs"
| rex field=_raw "<TimeCreated SystemTime='(?<TimeCreated>[^']
+)' "
| table TimeCreated
```

Time range: All time

✓ 8 events (before 6/24/26)

Job ▾

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⬇

Policy-Based Pool ▾

12:07:36.000 AM

No Event Sampling ▾

Events

Patterns

Statistics (8)

Visualization

Show: 20 Per Page ▾

✎ Format ▾

☒ Preview: On

TimeCreated ▲

2022-08-31T17:34:44.753560200Z
2022-08-31T17:38:40.513106600Z
2022-08-31T17:38:52.835733900Z
2022-08-31T17:39:49.241330600Z
2022-08-31T17:41:57.268855100Z
2022-08-31T17:43:19.090057300Z
2022-08-31T17:46:58.657885900Z
2022-08-31T17:47:04.788786400Z

The 8 executions occurred between 17:34 and 17:47 on the same day. The short time interval and high frequency suggest automated or scripted activity.

```
source="4688_windows-security.log" sourcetype="windowsseclogs"
| rex field=_raw "<TimeCreated SystemTime='(?[^\']+)'"
| table TimeCreated
```

As a result, this confirms repeated execution activity during a very short period of time.

Evidence Found

1. Eight EventCode 4688 events were identified in the Windows Security log. All events show execution of ntdsutil.exe on the Domain Controller.
2. The command line used was:
"ntdsutil "ac i ntds" "ifm" "create full C:\Windows\Temp\ntds_T1003" q q"
This command is used to export the Active Directory database (NTDS.dit).
3. The export destination, C:\Windows\Temp\ntds_T1003, is not a standard backup location.

The executing user was "Administrator", a Domain Admin account.
This is consistent with the privilege level required to perform this operation.

Based on the Windows Security logs, evidence strongly suggests MITRE ATT&CK T1003.003 (NTDS Credential Dumping) activity. Eight Event ID 4688 events showed repeated execution of ntdsutil.exe by the Administrator account. The command used exported the Active Directory database, and the activity occurred multiple times within a short time period. These findings are consistent with an attempt to access NTDS.dit and obtain Active Directory credential data.